Self 12 Bayes' rule & Total prob for RVs

$P(A \cap B) = P(A)P(B|A)$
$= P(B)P(A|B)$

Partition: $A_1, A_2$

$P(B) = P(B|A_1)P(A_1) + P(B|A_2)P(A_2)$

Discrete:

≈ Same as events

$X, Y \sim$ joint PMF $p_{X,Y}$

$$p_{X,Y}(x,y) = p_X(x)\, p_{Y|X=x}(y) = p_Y(y)\, p_{X|Y=y}(x)$$

$A: X = x \qquad B: Y = y$

$$p_X(x) = \sum_y p_{X,Y}(x,y) = \sum_y p_{X|Y=y}(x) \cdot p_Y(y)$$

$B: X = x$ Partition: $\bigsqcup_y (Y = y)$

Jointly continuous

$X, Y \sim$ joint PDF $f_{X,Y}$

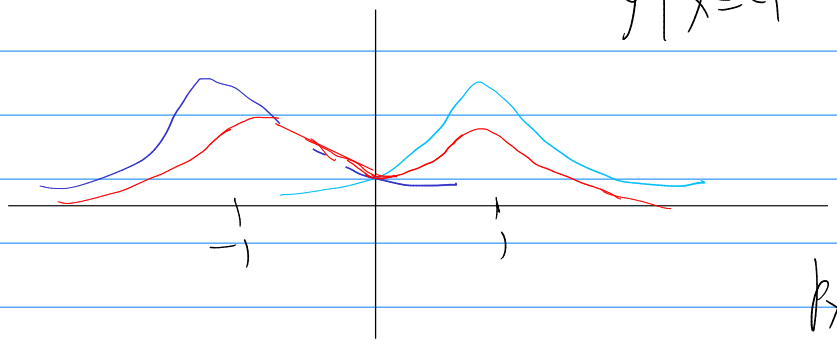
$$f_{X,Y}(x,y) = f_X(x)\, f_{Y|X=x}(y) = f_Y(y)\, f_{X|Y=y}(x)$$

$$f_X(x) = \int_{y=-\infty}^{\infty} f_{X,Y}(x,y)\,dy = \int_{y=-\infty}^{\infty} f_{X|Y=y}(x)\, f_Y(y)\,dy$$

Ex: $X \sim \{-1, 1\}$ (1/2, 1/2)

$Y|X=1 \sim$ PDF $\frac{1}{\sqrt{\pi}} e^{\frac{-(y-1)^2}{2}}$

$Y|X=-1 \sim$ PDF $\frac{1}{\sqrt{\pi}} e^{\frac{-(y+1)^2}{2}}$




"$X \cdot Y|X = Y \cdot X|Y$"

$$p_X(x) \cdot f_{Y|X=x}(y) = f_Y(y)\, p_{X|Y=y}(x)$$

$$f_Y(y) = f_{Y|X=1}(y)\cdot p_X(1) + f_{Y|X=-1}(y)\, p_X(-1)$$

$$= \frac{1}{\sqrt{2\pi}} e^{\frac{-(y+1)}{2}} \cdot \left(\frac{e^{y}+e^{-y}}{2}\right)$$

$$P_r(X=1 \mid Y=y) = p_{X|Y=y}(1) = \frac{\frac{1}{2}\cdot\frac{1}{\sqrt{2\pi}} e^{\frac{-(y-1)^2}{2}}}{f_Y(y)}$$ ← greater when $y > 0$

$$P_r(X=-1 \mid Y=y) = p_{X|Y=y}(-1) = \frac{\frac{1}{2}\cdot\frac{1}{\sqrt{2\pi}} e^{\frac{-(y+1)^2}{2}}}{f_Y(y)}$$ ← greater when $y < 0$

Ex: $P \sim \text{Uniform}[0,1]$

Given $P=p$, toss a coin with $P_r(H)=p$ indep. $n$ times

$X$ = # of heads

$$X \mid P=p \sim \text{Binomial}(n,p)$$

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$$f_P(p)\cdot p_{X|P=p}(k) = p_X(k)\cdot f_{P|X=k}(p)$$

$$f_P(p)\binom{n}{k} p^k (1-p)^{n-k} = p_X(k)\cdot f_{P|X=k}(p)$$

$$p_X(k) = \int_0^1 \binom{n}{k} p^k (1-p)^{n-k} f_P(p)\, dp = \frac{1}{n+1}$$

$f_{P|X=k}$: valid PDF

# Multivariate distributions

$X = (X_1, X_2, \ldots, X_n)$ : jointly distributed

$X^n$ (or) $\underline{X}$ (or) $\overline{X}$

## All discrete:

Joint PMF $p_{X_1 \cdots X_n}(\underbrace{x_1, \ldots, x_n}_{x}) = P(X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n)$

$X_i \in \mathcal{X}_i$ ($\mathcal{X}_i$: discrete) $\quad x_i \in \mathcal{X}_i$

$(x_1, \ldots, x_n) \in \mathcal{X}_1 \times \cdots \times \mathcal{X}_n$

$S \subseteq [n] \triangleq \{1, 2, \ldots, n\}$

$X_S = (X_i : i \in S)$ sub-vector

$x_S = (x_i : i \in S)$

$p_{X_S}(x_S) = \sum_{x_{S^c}} p_X(x)$